

Y3S2 Group Project Explaining the Code

Jack D. Maher (jxm1517)

February 2026

1 Pre-processing

1.1 Atmospheric Assumptions and Composition

1.1.1 One-Dimensional Isothermal Atmosphere

When forming a model absorption spectrum, our script takes the input planetary parameters and assumes a one-dimensional, isothermal atmosphere. The one-dimensional approximation neglects latitudinal and longitudinal variations, treating the atmosphere as spherically symmetric. This means we do not see the effects of day-night temperature contrasts or banding that can exist in real exoplanet atmospheres.

The assumption of a one-dimensional atmosphere is common in exoplanet atmospheric modeling [1, 2]. and is a reasonable approximation when the goal is to understand general trends in the spectrum.

1.1.2 Ideal Gas Assumption

The atmosphere is modeled as an ideal gas. Under this assumption, the number density (n) is calculated from the ideal gas law as $n = \frac{P}{k_B T}$ where k_B is the Boltzmann constant and T is the temperature.

This assumption is valid for exoplanet atmospheres at the pressures and temperatures typically probed in transmission spectroscopy. The ideal gas law allows us to directly relate pressure measurements to particle number densities, which are essential for calculating opacity and optical depth.

Under assumptions 1.1.1 and 1.1.2, the pressure (P) decreases with radial distance (r) according to $P(r) \propto \frac{1}{r^2}$, reflecting how pressure falls off in a hydrostatic atmosphere moving outward from the planet.

1.1.3 Constant Composition

The atmosphere is also assumed to have a uniform mixing ratio for each molecules in the atmosphere. This means the atmosphere's composition does not vary with altitude or pressure, although the density of the gasses will decrease with altitude as discussed in Section 1.1.2. In reality, atmospheric composition can vary with height due to diffusive separation and chemical processes [3], but this approximation is reasonable and is a standard assumption used in other exoplanet atmospheric studies [4]. Having a constant composition simplifies the calculation significantly, allowing us to specify the atmosphere's composition once and apply it uniformly across all pressure levels.

1.1.4 Cloud Deck

The cloud deck (at a specified pressure P_{cloud}) is treated as a hard cutoff and is assumed to be 100% opaque below that level. This represents an idealized cloud deck where all light is completely blocked below a sharp pressure boundary. This assumption effectively sets an inner radius r_{cloud} below which no light is transmitted.

1.2 Radiative Transfer and Optical Depth

When the planet passes in front of the star, some light rays pass through the upper atmosphere while deeper rays are blocked by the opaque interior (clouds or solid surface). The aim of this step is to calculate how much starlight is removed along each ray at each wavelength.

For transmission spectroscopy we track attenuation of the incoming starlight, which is written as

$$I_\lambda(s) = I_{\lambda,0} e^{-\tau_\lambda(s)} \quad (1)$$

where $I_{\lambda,0}$ is the incoming specific intensity from the star, s is the distance along the ray, and τ_λ is the optical depth.

Therefore, if $\tau_\lambda \ll 1$ the atmosphere is mostly transparent at that wavelength and if $\tau_\lambda \gg 1$ the ray is mostly blocked.

The script then builds the wavelength-dependent optical depth from small path elements using the relation in equation 2.

$$d\tau_\lambda = \kappa_\lambda \rho ds = \sum_i n_i \sigma_{\lambda,i} ds \quad (2)$$

where κ_λ is the mass absorption coefficient, ρ is the density, n_i is the number density of species i , and $\sigma_{\lambda,i}$ is its absorption cross section.

In other words, each layer contributes an amount of absorption set by how many particles are present and how strongly they absorb at that wavelength. Summing these contributions along the full ray gives τ_λ for that impact parameter. This is the key link between atmospheric structure/composition and the observed spectrum: wavelengths with larger optical depth higher in the atmosphere produce a larger effective transit radius and therefore a deeper transit signal.

1.3 Pressure Grid and Hydrostatic Equilibrium

The code represents the atmosphere on a 1D pressure grid from $P = 10^{-4}$ Pa to $P = 10^7$ Pa. For each pressure level P_j it assigns a temperature $T_j = T_p P_j$ and number density $n_j = \frac{P_j}{k_B T_j}$.

To know how far rays travel through each layer, we need to find the radius r at each pressure level. This is obtained by integrating the equation of hydrostatic equilibrium, $\frac{dP}{dr}$, which leads to the expression in equation 3.

$$dr = -\frac{k_B T}{\mu g} d \ln P \quad (3)$$

The calculation of gravity relative to radius can be stated as $g(r) = g_{\text{ref}} \left(\frac{r_{\text{ref}}}{r} \right)^2$, where g_{ref} is the surface gravity at a reference pressure P_{ref} and $r_{\text{ref}} = R_p$ is the corresponding radius. From this reference point, the script steps outwards and inwards in even steps of $\ln P$ to build the arrays $r(P_j)$ and $g(P_j)$. This converts the original pressure grid into a grid of atmospheric shells.

We then set the composition of the atmosphere in terms of logarithmic mixing ratios ($\log_{10} X_i$) for each molecule. This was calculated as a typical sub-neptune's composition [INSERT REFERENCE]. From these mixing ratios, the mean molecular weight is calculated as $\mu = \sum_i X_i m_i$.

The absorption cross sections ($\sigma_{\lambda,i}(P, T)$) are loaded as arrays and we use RegularGridInterpolator to evaluate the cross sections at the pressures and temperatures of the model atmosphere and at the wavelengths of interest. For each wavelength λ_k and layer j , the total absorption per unit path length is then calculated as $\sum_i n_{i,j} \sigma_{\lambda_k,i}(P_j, T_j) = n_j \sum_i X_i \sigma_{\lambda_k,i}(P_j, T_j)$.

1.4 Collision-Induced Absorption

We then also account for collision-induced absorption (CIA) from $\text{H}_2\text{-H}_2$ and He-H_2 pairs. CIA occurs when two colliding molecules form an interaction with an induced dipole moment, allowing transitions to absorb radiation. This is important in H_2 -rich atmospheres because H_2 has no permanent dipole and therefore weak intrinsic rotational-vibrational line absorption on its own. Unlike narrow molecular line features, CIA typically contributes a broad continuum-like opacity that fills spectral windows between molecular bands. [INSERT REFERENCE FOR CIA CROSS SECTIONS] Because CIA is a two-body process, the contribution to τ_λ scales as n_i^2 rather than n_i .

$$\tau_\lambda^{\text{CIA}} \propto n_j^2 X_{\text{H}_2}^2 \sigma_\lambda^{\text{H}_2\text{-H}_2} + n_j^2 X_{\text{H}_2} X_{\text{He}} \sigma_\lambda^{\text{He-H}_2} \quad (4)$$

After these contributions are combined, the code holds a 2D array `opacity[k,j]` which represents $\sum_i n_{i,j} \sigma_{\lambda_k,i}$ for each wavelength and pressure level.

1.5 Slant Optical Depth and Transit Integration

Given the radial grid r_j and the opacity at each layer, the next part of our script finds the optical depth $\tau_\lambda(r)$ along rays that pass through the planet at different layers r . For a ray with minimum radius r_i , the path length through layers at larger radii is set by equation 5 so that the path length across a shell between r_j and r_{j+1} is $\Delta s_j = s(r_{j+1}) - s(r_j)$.

$$s(r) = \sqrt{r^2 - r_i^2} \quad (5)$$

The code loops over the layers above each impact parameter and approximates the slant optical depth by the sum 6 where the opacity is averaged between adjacent layers to obtain an estimate.

$$\tau_\lambda(r_i) \approx \sum_j \left[\frac{\text{opacity}_{k,j} + \text{opacity}_{k,j+1}}{2} \right] \Delta s_j \quad (6)$$

If the pressure at a given radius exceeds the prescribed cloud-top pressure P_{cloud} , a very large additional optical depth (e.g. $\tau \sim 10^3$) is added to mimic an effectively opaque cloud deck.

The fraction of light transmitted along that ray is $e^{-\tau_\lambda(r_i)}$. The script converts this to an effective transit radius with the standard annulus-integration formula for transmission spectroscopy in equation 7.

$$\Delta_\lambda = \frac{\pi R_p^2 + 2 \int_{r>R_p} r [1 - e^{-\tau_\lambda(r)}] dr - 2 \int_{r<R_p} r e^{-\tau_\lambda(r)} dr}{\pi R_s^2}. \quad (7)$$

The first term represents the area of an opaque disk of radius R_p . The second term adds on the contribution from atmospheric annuli above this radius, weighted by the fraction of light they block. The third term subtracts light that is able to leak through layers that are geometrically inside R_p but not completely opaque because of the finite optical depth. In practice these integrals are again evaluated numerically using the discrete r_i grid and the precomputed values of $e^{-\tau_\lambda(r_i)}$.

The result is a model transit depth Δ_λ as a function of wavelength, which the notebook plots in parts-per-million (ppm). Changing the planetary radius, mass, temperature, mean molecular weight, or cloud-top pressure directly affects the computed scale height and optical depths, and therefore changes the amplitude and shape of the spectral features.

References

- [1] S Seager and D D Sasselov. Theoretical transmission spectra during extrasolar giant planet transits. *Astrophys. J.*, 537(2):916–921, July 2000. doi: 10.1086/309088.
- [2] Timothy M Brown. Transmission spectra as diagnostics of extrasolar giant planet atmospheres. *Astrophys. J.*, 553(2):1006–1026, June 2001. doi: 10.1086/320950.
- [3] Julianne I Moses, C Visscher, J J Fortney, A P Showman, N K Lewis, C A Griffith, S J Klippenstein, M Shabram, A J Friedson, M S Marley, and R S Freedman. DISEQUILIBRIUM CARBON, OXYGEN, AND NITROGEN CHEMISTRY IN THE ATMOSPHERES OF HD 189733b AND HD 209458b. *Astrophys. J.*, 737(1):15, August 2011.
- [4] N Madhusudhan and S Seager. A temperature and abundance retrieval method for exoplanet atmospheres. *Astrophys. J.*, 707(1):24–39, December 2009.